BAS

Ballistics and Scoring

User Guide

Version 1.17.0 (build 27) · August 2026
For Android 8.0 and later

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Contents

Safety first

> **Range safety**
>
> Operate the app only from behind the firing line, with the firearm made safe and the range in a condition permitted by the range officer.
>
> Never handle a phone and a loaded firearm at the same time.

Nothing in this guide overrides the range officer. If scoring a string means walking forward, changing your position, or taking your attention off the firing point, do it when the range is cold and not before. The app is designed so that every step that needs your hands can wait.

If you are shooting prone, set the phone up before the relay begins and leave it alone. Range mode, spoken corrections and the remote trigger all exist so that you do not have to reach for it — see section 11.

1. What BAS does

BAS carries a shooter through the whole of a session, and it does two jobs in the order they are actually needed.

First it puts the shots on centre. You record the shot through a phone on the scope, a scope-mounted camera or a digital scope, and BAS measures the crosswind acting on the bullet from the video itself, then converts that wind — together with your zero, your load and the air conditions — into the clicks to dial on your turrets.

Then it grades the group. You register the target face once, and every hole is found, measured against the ring geometry of the selected face, and scored under the rules of the selected match. BAS reports the running total, the group centre and dispersion, and the sight correction needed to move the group onto the point of aim.

The two halves share one description of your equipment, so the rifle, load and sight you enter once drive both the trajectory solution and the scoring gauge.

1.1 How the wind is measured

BAS measures the wind three ways, depending on what is visible in your video:

- Vapour trail — every bullet leaves a faint trail of disturbed air. BAS tracks how that trail drifts downwind after the shot. This is the general method, and it works for rimfire and centrefire.
- Tracer — for tracer ammunition, BAS tracks the burning element in flight and reads the wind from how far it is pushed sideways.
- Projectile — for a visible projectile (an airgun slug, or a bullet filmed through a magnified digital scope) BAS tracks the projectile itself.

1.2 Disciplines covered

BAS is built for long range: centrefire rifle in .308, 6.5 Creedmoor, .223 and similar chamberings out to 1000 yards, where the wind estimate and the scope correction earn their keep. Rimfire rifle and pistol at 25 m and 50 m, and air rifle and air pistol at 10 m, are supported for the scoring side, with the target faces and rule sets those disciplines need.

> **Before you rely on a score**
>
> BAS is a training and practice aid. Official match results are determined by the appointed jury under the current rulebook of the governing body.
>
> Ring dimensions and rule parameters shipped with the app are nominal published figures and are editable — verify them against the rulebook in force for your competition.

2. What you need

- An Android phone running Android 8.0 or newer.
- For the ballistics side: a rigid mount that holds the phone with its camera looking through, or squarely aligned with, your scope — or a scope-mounted camera that streams or records (section 8).

- For the scoring side: a view of the target, either from the phone on a stand or from a photograph of the card taken afterwards.
- Permissions, which BAS asks for the first time each is needed: Camera and Microphone (to record and to detect the shot), Nearby devices (for a Kestrel or a rangefinder), and access to your files.
- Optional: a Kestrel weather meter for live air data (section 9), a rangefinder (section 10), and a chronograph reading of your load's muzzle velocity for best accuracy.

BAS runs entirely on your phone. It has no account and sends none of your data anywhere, apart from the optional AI scoring service described in section 12, which you enable and key yourself.

3. The screens

Along the bottom are the main tabs, in the order the work actually flows. Move between them by tapping, or by swiping left and right.

Tab	What it is for
Home	The active setup at a glance — profile set, firearm, load, sight, target, rules and distance — and the way into Range mode.
Ballistics	Recording or importing the video, the shot conditions, and the camera.
Scoring	Registering the target and scoring the holes.
Results	Both results, with a Scoring / Ballistics switch at the top.
Settings	Equipment, cameras, rangefinder, targets, rules, display and backup.

Target faces, competition rules, the diagnostic log and backup each have their own section in Settings. The version line is at the foot of Settings — quote it if you ever report a problem.

4. Setting up your equipment

Accuracy here determines the accuracy of everything else. Work through it once; everything is remembered between sessions.

4.1 Profile sets — the quick way

A profile set is a complete rig: firearm, load and sight together. BAS ships with ready-made sets and applies one on first launch, so the app is usable before you have entered anything. The default is the Rimfire trainer — Ruger Precision Rimfire .22LR, Federal Champion 40 gr, Vector Optics Continental 5-30x56.

Open Settings and tap Profile sets. Opening it also opens Firearm, Load and Sight, because those three are what a set is made of. Pick a set, tap Apply, and every screen follows it.

4.2 The load

Calibre, weight, muzzle velocity and the G1 ballistic coefficient, as printed on the box. Two further fields matter if you shoot across seasons: the muzzle-velocity temperature coefficient and its reference temperature, which let BAS correct the velocity for the day.

Enter your chronographed muzzle velocity if you have one. It is the single most valuable number on this screen.

4.3 The firearm

Barrel length, twist rate, sight height over the bore, and the zero distance. The zero distance drives the whole trajectory solution, so set it to the range you actually zeroed at — the Distance button on the Ballistics tab can edit and save it directly.

4.4 The sight

Click unit, travel, magnification and the sight's height above the bore. BAS supports quarter-MOA, eighth-MOA, half-MOA, tenth-MRAD and hundredth-MRAD turrets, sights specified in millimetres at a reference distance — most match iron sights — and sights with no adjustment at all, where the correction is reported as the movement the point of impact needs.

5. Ballistics — measuring the wind and dialling on

5.1 Recording a shot

Open the Ballistics tab and set the conditions for the shot.

Control	What it does
Target distance	The distance to your target. This drives the whole solution, so set it accurately — see section 10.
Camera zoom	Magnifies the live view. More magnification gives a better wind measurement, as long as the target stays in frame.
FOV and Zoom	The camera's field of view and the magnification of the clip. Filled automatically when you record in the app; set by hand only for imported video.
Frame rate	Only the rates your phone genuinely supports are offered; higher rates give the analysis more to work with.
Shot break	A manual fallback for the shot instant. Normally leave it — BAS finds the moment from the sound of the shot.

There are four ways to get the clip:

- Record — frame the target, let the picture settle, then record. BAS locks exposure and focus the instant recording begins and switches off image stabilisation, because both would corrupt the measurement.
- Arm (auto-trigger) — recording starts by itself when the microphone hears the shot.
- Import — analyse a video already on the phone, including a clip from a digital scope's memory card.
- From the camera — download the newest clip over the camera's own Wi-Fi, or let Auto-collect fetch and analyse it the moment you stop recording (section 8).

Then tap Analyze trail. The analysis takes a few seconds and moves you to the results.

5.2 Reading the result

Row	What it means
Wind	The crosswind for this shot, with its spread and a direction arrow. The ± figure is how much the wind varied during the measurement.
Session	From the second shot onward, the average crosswind over the last ten minutes. In steady conditions this is the number to dial.
Impact / Correction	Where the shot would land uncorrected, and the clicks to dial — with the same correction in MRAD and MOA beside it.
Confidence	How much trust to place in this measurement. Faint trails at short range give low confidence; that is normal. Treat a low-confidence single shot as a hint and lean on the session average.

5.3 The chart

The chart plots the crosswind BAS measured. While the bullet is still in flight the horizontal axis is distance from the muzzle, up to your target distance. Because the vapour trail keeps drifting for a few seconds after the bullet has arrived — and those later moments still measure the wind — the chart switches to time after the shot once the bullet has reached the target. There you are looking at how the wind behaved over those seconds at your firing position, which is the picture behind the ± figure.

5.4 Calibrating for your rifle

Two calibrations make the results match your actual rifle and load. Do them in this order.

- Drop calibration aligns the elevation with your load's real trajectory. Enter one point from the manufacturer's drop table — a distance and its drop — and BAS adjusts its model to reproduce it, then shows the resulting table to check. Redo it whenever you change the ballistic coefficient or muzzle velocity.
- Wind calibration pins the wind measurement to a known reference. Fire an analysed shot while you measure the actual crosswind, enter that value, and tap Solve. Tracer and projectile measurements are geometrically exact and do not need this.

5.5 Air rifles

For an airgun, tick the airgun projectile option on the load. BAS then tracks the pellet or slug itself rather than a vapour trail, which a sub-sonic projectile does not leave. Everything downstream is unchanged.

6. Scoring a target

There are two routes, and either can be used with the same card.

- From a photograph. Photograph the card after the string, load it, and score it. This is the more accurate route and the one to learn first.
- Live, as you shoot. The phone watches the target on a stand and adds each shot as it appears. Convenient, but it depends on a steady view and good light.

6.1 From a photograph

Photograph the card square-on and fill the frame with it. Even light matters more than bright light: a hard shadow across the card is the commonest cause of a missed hole.

On the Scoring tab choose Upload a target photo to score, or take the newest photograph straight from a connected GoPro. Then register the target: Identify target and register fits the printed rings and takes the scale from their spacing, which is more accurate than the aiming mark alone; Auto-detect uses the face you have already selected. If the card was photographed at an angle, register by tapping its four corners instead.

Supplying a second photograph of the same card BEFORE it was shot is worth the trouble. Scoring then becomes a difference, and the printed rings, the paper grain and the black aiming mark all cancel because they appear in both images.

6.2 Live from the camera

Put the phone on a stand with the target in frame, capture a clean reference frame of the unshot card, and start live detection. Each new hole is found by differencing against the reference, scored, and added to the running total.

The ring guide draws the selected face's own rings over the preview. Lining them up does more than centre the picture: the circles are at that face's true proportions, so a card whose rings are spaced differently will not line up however far you move — which is the quickest way to catch a wrong target face, and a wrong face is the commonest reason nothing is detected.

6.3 Reading the Results screen

The plot shows the shots on the selected face, with the group centre and the point of aim. My photo switches the background to your own rectified card, which is the only view in which a missed shot is visible.

Below it are the total, the shot distribution, the group centre and dispersion, and the correction. The distribution is worth reading: 95 as ten 9s and 10s is a shooter who needs a sight correction, while the same 95 as eight 10s and two 7s is a shooter throwing the occasional flyer. One is fixed with a turret and the other with the shot process, and the total alone cannot tell them apart.

Anything the detector got wrong can be corrected by hand — tap the plot to add or remove a shot. A hand-placed shot is recorded as such.

7. Target faces and rule sets

A target face is the geometry — the ring radii, the aiming mark, the card size. A rule set is the convention — the gauge diameter, decimal or integer scoring, the number of shots, whether inner tens are counted. Both ship with the app, both are editable, and both have their own section in Settings.

Choose the face you are actually shooting. If the printed rings will not line up with the guide, the face is wrong — no amount of registration will fix that.

8. Cameras and video sources

Both the Ballistics and Scoring tabs carry a Camera button and a Configure button beside it. Configure only ever offers what the selected camera supports.

Camera	What BAS can do with it
Phone camera	Live preview, recording, single frames. Nothing to configure.
GoPro (HERO9 and later)	Start and stop recording, digital zoom, load a preset, camera state, keep awake, download the newest clip, and a live preview. Uses the published Open GoPro interface.
TACTACAM / ShotKam	Download the newest file over the camera's own Wi-Fi, and a discovery scan that reports what the camera exposes.
RTSP / MJPEG stream	A digital scope or IP camera by stream address.

8.1 Working over a camera's Wi-Fi

A camera's own access point has no internet, so join it in the phone's Wi-Fi settings first. BAS binds to that network deliberately while it talks to the camera; without that, requests would leave over mobile data and the camera would be unreachable. Auto-reconnect, in Range options, makes BAS wait longer for a camera that is slow to appear.

8.2 Auto-collect

Arm Auto-collect and BAS stands by, watching the camera for a new clip. Stop recording — with the camera's own remote if it has one — and the clip is downloaded and analysed with no further interaction. It ignores the clip that was already there when you armed it, so it only ever acts on a genuinely new recording.

8.3 Camera settings

Settings records how a Wi-Fi camera is set: zoom, video size, white balance, exposure compensation, mains frequency, red dot and stabilisation. Two of these change what BAS does rather than merely being noted. With the red dot declared on, detections within half a gauge of the frame centre are dropped, because the dot is drawn into the picture exactly where the ten ring usually sits. And the arriving stream size is checked against what you declared, with the usual reason for a mismatch explained.

Turn stabilisation off on any camera used for trail analysis. It moves the picture between frames, which is precisely what the measurement assumes does not happen.

9. Weather and the Kestrel

Air conditions affect the trajectory, so BAS uses the best data it has. Your phone's barometer supplies pressure. For temperature and humidity — which most phones cannot sense — connect a Kestrel weather meter over Bluetooth.

- The Kestrel 5700 Elite (and the Ruger-branded version) with LiNK pairs normally in your Android Bluetooth settings.
- The Kestrel DROP D3 does not appear in the paired list — that is normal. BAS finds it by scanning when you tap Kestrel.

Switch the meter on nearby and tap Kestrel. The status line shows each value and its source, and the reading is remembered between sessions with its age noted. As a cross-check, the Kestrel's pressure and your phone's should agree to about 1 hPa.

10. Distance and rangefinders

The distance to the target drives the whole solution. BAS offers three ways to obtain it, from the Distance button on both tabs and from Settings under Rangefinder and distance.

Route	How it works
Through a Kestrel 5700 Elite	Leica, Vortex and SIG BDX-X units push their range into the Kestrel, and BAS reads it from there — one link covers three brands.

Route	How it works
Direct to the rangefinder	SIG KILO, Leica Geovid Pro and CRF .COM, Vortex Fury HD and Razor HD, Vectronix Terrapin-X, Tangoinnos FIRE4000, or a generic Bluetooth unit.
By hand	Type it, or take the rig's zero (calibration) distance — the range the scope, and usually the rangefinder, was set up at. Always available.

10.1 Teaching BAS the range signal

No rangefinder maker publishes its Bluetooth protocol, so BAS listens to the device and decodes what looks like a range. Because a temperature or a battery level can look like one too, the first reading from a device is offered for confirmation: check it against the rangefinder's own display and accept it. BAS then remembers exactly which signal and which unit to read, and every later range comes from that with no guessing.

If the wrong value was confirmed, Forget the learned range signal clears it and the next reading asks again.

11. Range mode and hands-free use

Most of a session is spent prone, where reaching for the phone costs the position. These options exist to remove that, and all are in Settings under Range options.

Feature	What it does
Range mode	A full-screen glance, opened from Home: the wind correction in the largest type, the group correction and the score below, each with the angle in MRAD and MOA. Readable at arm's length.
Speak corrections	Announces the shot and the correction — “Come up 3, left 1.” Off until you enable it.
Keep screen on	Stops the phone sleeping mid-string. On by default.
Remote trigger	The volume buttons or a Bluetooth shutter fire the screen's main action — analyse on Ballistics, score now on Scoring.
Auto-collect	Fetches and analyses a new clip as soon as recording stops (section 8.2).
Skip confirmations	Turns the clear-shots and remove-marks prompts into immediate actions.

The last correction is also drawn over the live viewfinder on both tabs, so it can be read while you are still behind the scope.

12. Scoring with an AI service

BAS can send a rectified photograph of the card to an AI service and use its answer as a second opinion, or to find and score the shots outright. This is optional, off by default, and uses your own API key.

The app's own algorithms remain the more precise of the two: BAS measures a hole it can see to within a millimetre or two, where a vision model places it to several millimetres. What the model does better is counting. Over-detection — printing read as a shot — is the app's measured failure mode, and how many holes are on a card is a counting question.

Registration always stays with the app under either option. Without knowing where the card is and how big it is there is no millimetre grid, and without a grid nothing can be drawn in the right place.

13. Display, backup and other settings

Setting	Options and notes
Units	Metric (m, mm, cm) or Imperial (yd, in). Metric is the default.
Theme	Dark, Day, Night — Green, Night — Red. Dark is the default and saves power on OLED screens.
Viewfinder reticle	None, crosshair, duplex, mil-dot, MOA grid, German #4, circle and dot, MOA and MRAD wind trees, or your own image.
Full screen	Hides the system bars while shooting.
Diagnostic log	Records every analysis, camera setting, stream handshake and Bluetooth exchange.
Backup and reset	Export everything to a file, restore it, or reset the active profiles.

The night themes exist to preserve dark adaptation. Under Night — Red the app keeps every element red, including the reticle, so nothing on screen costs you your night vision.

14. Getting good results

- Calibrate drop first, then wind.
- Use the highest zoom that still keeps the target framed.
- Let the exposure settle on the target before you record.
- Turn image stabilisation off — on the phone BAS does this for you; on a scope camera, set it off and declare it in camera settings.
- Shoot a string and read the Session row: several shots give a far tighter figure than one.
- Enter your chronographed muzzle velocity, and its temperature coefficient if you shoot across seasons.
- For scoring, photograph the card square-on in even light, and supply a clean 'before' photograph when you can.
- Treat low-confidence single shots with caution; lean on the session average in steady wind.

15. Accuracy: what to expect, and what to check

This section is here because an app that hides its limits is worse than one that has none. Read it once.

15.1 How the scoring is checked

A photographed ISSF 10 m Air Pistol card is built into the app and scored on every build. Its answer was established by hand, independently of the app's own code. It carries five scoring shots — a 9, a 6, a 2 and two 1s, nineteen points — and two shots that missed the rings altogether. The app scores that card nineteen, at both full and half resolution, with no false marks. Printed cards punched at coordinates known to a tenth of a millimetre cover what a photograph cannot test.

15.2 What is still imperfect

- Shots that merge. Two holes about a gauge apart are separated correctly. Three or more in a rosette are found as one region and then not reported, because splitting along one axis is meaningless for a round cluster. A tight ten-shot group in the bullseye is the case handled worst.
- Marks outside the scoring rings, when you have asked to see shots that missed. Printing and genuine misses are interleaved there and nothing the app measures separates them. With the setting off — the default — no mark can appear outside the rings at all.
- Steeply angled photographs. Modest tilt is corrected well; beyond that the app is guessing.
- Wind estimates from a faint or short trail. The confidence figure says so; believe it.
- A rangefinder reading is only as good as the signal BAS was taught. Confirm the first one against the device's own display.

16. If something looks wrong

Symptom	What to do
The preview is black	Leave the tab and return — the camera rebinds on every visit. Check the camera permission.
No wind chart after an analysis	The solution was not valid for the rig: an air-rifle profile cannot reach a rifle target. Choose the profile set you are actually shooting.
Nothing is detected on the card	Usually the wrong target face. Line the ring guide up with the printing — if the rings cannot be made to match, the face is wrong.
A camera's Wi-Fi will not connect	Join the camera's network in the phone's Wi-Fi settings first, then try again. Turn on Auto-reconnect to wait longer.
The rangefinder gives a silly number	Use Forget the learned range signal, then confirm the correct reading next time.
The app closed unexpectedly	The next launch shows the recorded error with a Share button.
Anything else	Open the diagnostic log. Every step is recorded there, and it has Save and Share buttons.

The version line at the foot of Settings identifies your exact build — include it in any report.

17. A note on trust

BAS gives you a measurement, not a guarantee. Every wind estimate carries an uncertainty, and it is shown for a reason: in gusty conditions, or from a faint trail, the spread will be wide and the confidence low. Every score is measured from a hole in paper, and where a hole is marginal the gauge and the rule set decide — both of which are published figures you can check.

Read those figures, prefer averages over single shots, verify against your own manual corrections while you build confidence in the tool, and never let a number on a screen substitute for safe range practice and your own judgement.

BAS 1.17.0 — RFSAT